

Information Quality Brief History

John R. Talburt, PhD, IQCP

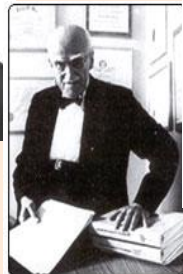
Quality Pioneers-Proven Principles

Manufacturing Quality
Pioneers

Japanese Quality
Pioneers

Information Quality
Pioneers

Walter Shewhart
W. Edward Deming
Joseph Juran
Armand Feigenbaum
Philip Crosby



Richard Wang
Larry English
Tom Redman
Danette McGilvray
Laura Sebastian-Coleman
Dave Loshin





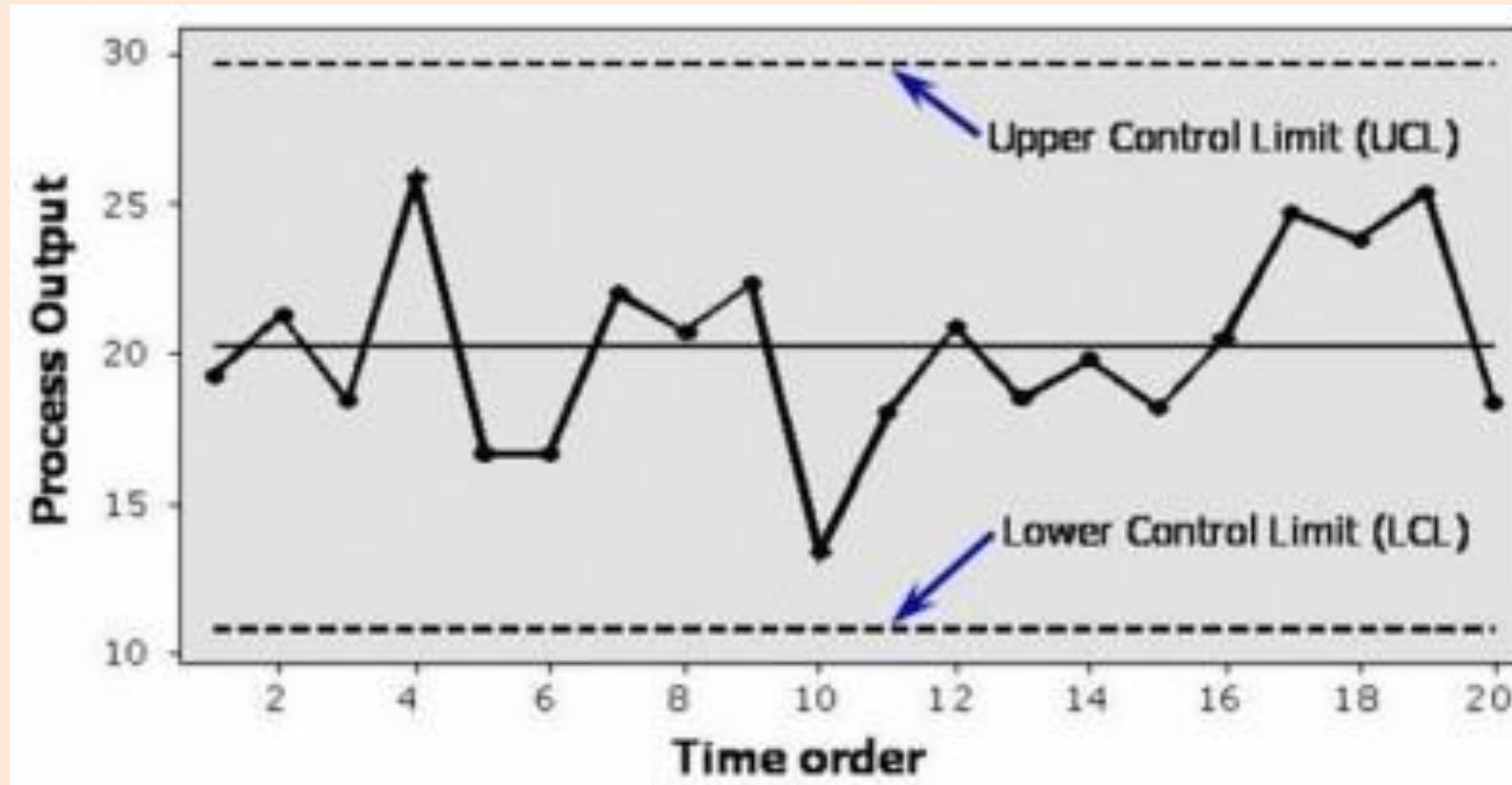
WALTER SHEWHART
1891 - 1967

Major Contributions

American Physicist, Engineer, and Statistician

- The “Father” of statistical process control
- Pioneered the application of statistics to quality
- Principles
 - Data have no meaning apart from their context
 - Data contain both signal and noise. To obtain information you must separate signal from noise
 - Inventor of Plan – Do – Check – Act cycle

Statistical Process Control





W. EDWARDS DEMING
1900 - 1993

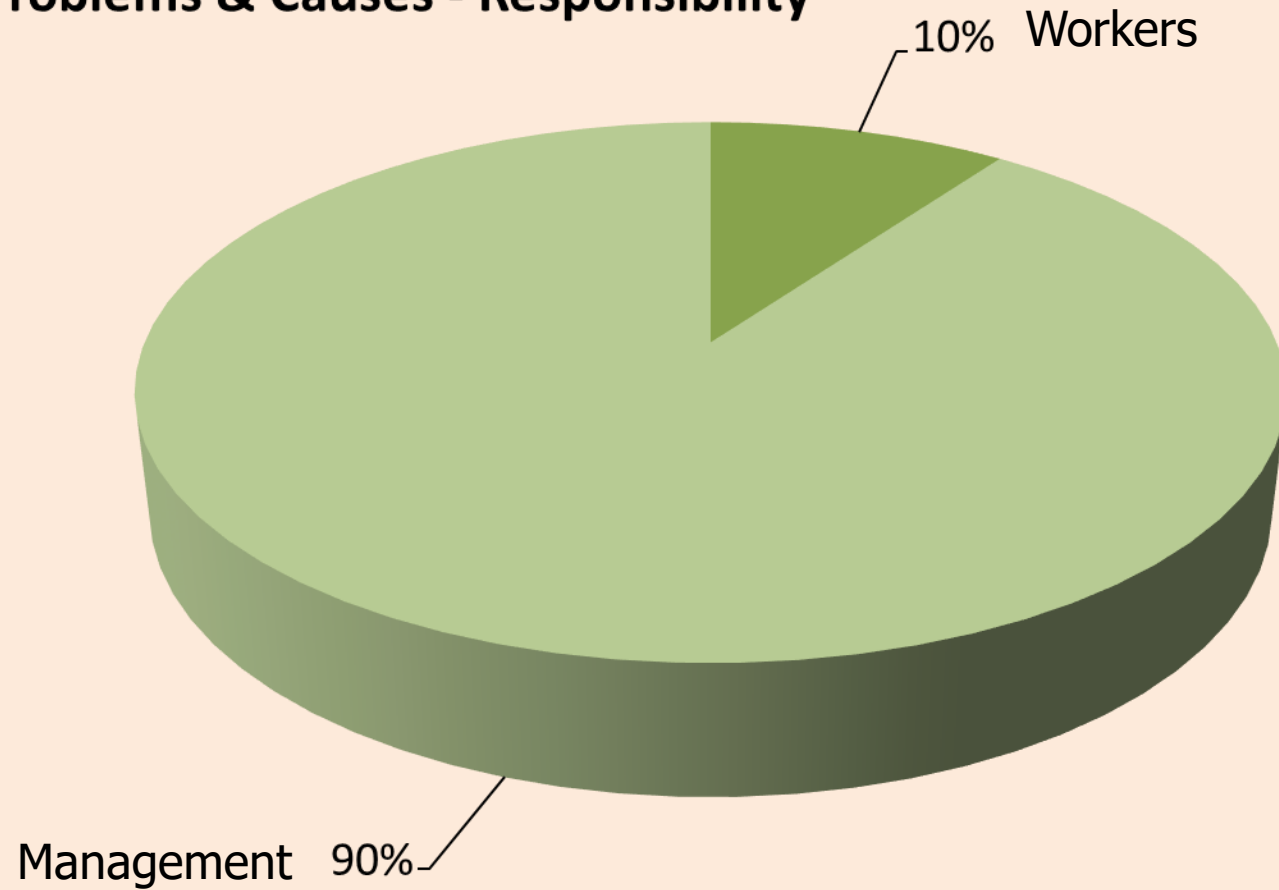
Summary of Deming's Work

American Engineer, Statistician, and Professor

- System of Profound Knowledge
- 14 Points for Organisational Transformation
- 7 Deadly Sins of Management
- Adopted Shewhart's PDCA as Plan – Do – Study – Act (PDSA)
- Advocate of Shewhart's Statistical Process Control
- The importance of Process thinking

Key Deming Message

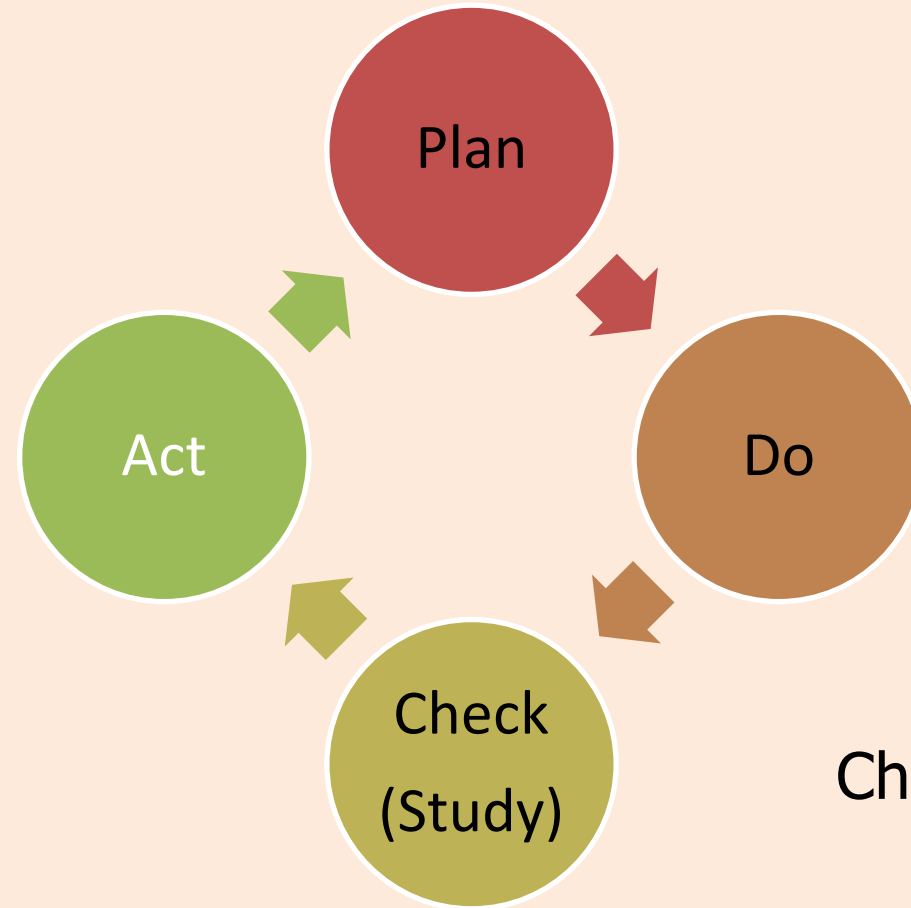
Problems & Causes - Responsibility



7 Deadly Sins (Symptoms)

- 1. Lack of Constancy of Purpose**
- 2. Emphasis on short-term profits**
- 3. Performance Appraisals that promote fear**
- 4. Mobility of Management**
- 5. Management by the use of visible numbers.**
- 6. Excessive medical costs**
- 7. Excessive cost of liability**

Continuously Improve the System of Production and Service



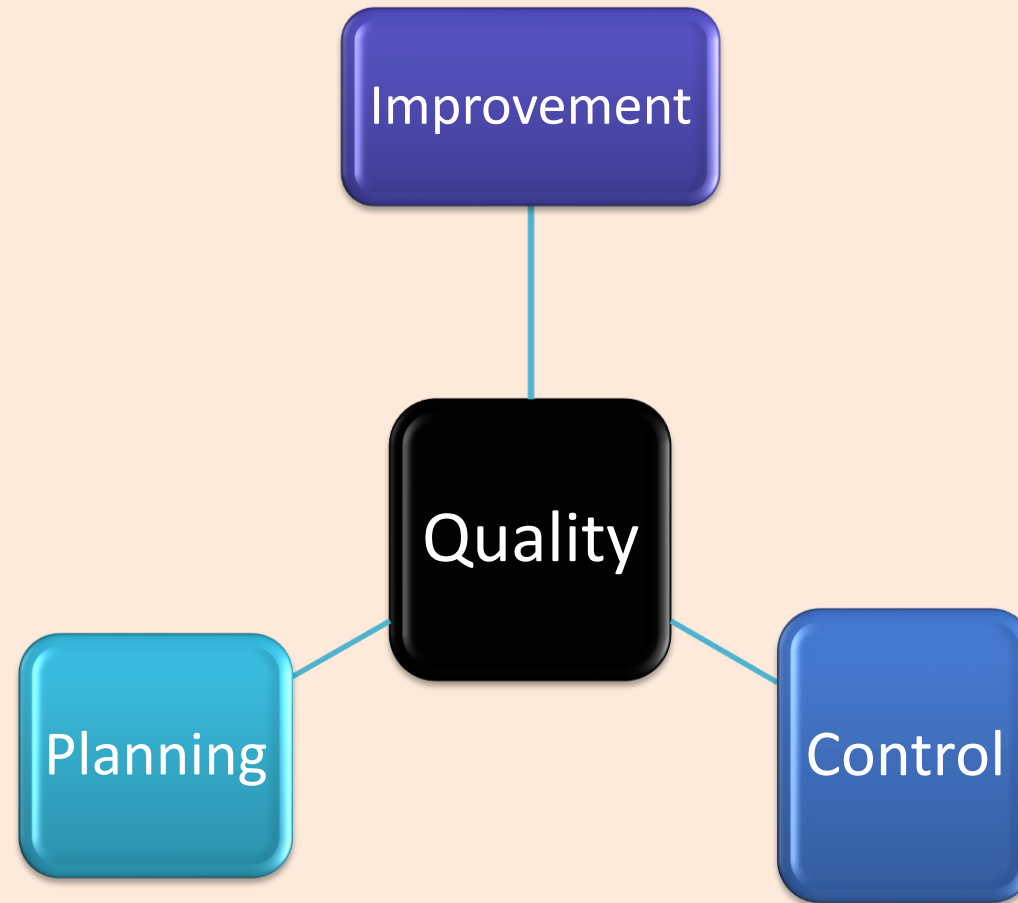
Change for the Better

KAIZEN



JOSEPH JURAN
1904 - 2008

The Quality Trilogy



Quality Planning

Who?

..are your customers?

What?

..are their needs and wants?

..are their quality characteristics?

How?

..will you make the product/service?

..will you deliver the product/service?

Quality Control

- Quality control, in Juran's view, is the governance process that makes sure things don't get worse
- Inspection of actual performance against planned performance
- Resolution of differences (root cause investigation)
- Makes use of Statistical measures of performance to ensure control

Quality Improvement

- Focus on fitness for use and error reduction
- Objective is a new level of performance superior to previous levels
- Juran advocates “**breakthrough thinking**” to change the system and improve performance



ARMAND FEIGENBAUM
1922 - 2014

Major Contributions

American Quality Control Expert

- Total Quality Control - Everyone working for full customer satisfaction
- “Hidden” plant - extra work performed in correcting mistakes hidden within any factory
- Accountability For Quality - quality must be visible at the highest levels of management
- Quality Costs – quantifying the cost of quality



PHILIP CROSBY
1926 – 2001

Major Contributions

American Businessman

- Published the book *Quality is Free* (1979)
- DIRFT system – “doing it right the first time”
 - Quality is conformance to requirements
 - The system of quality is prevention
 - The performance standard is zero defects
 - The measure of quality is the price nonconformance



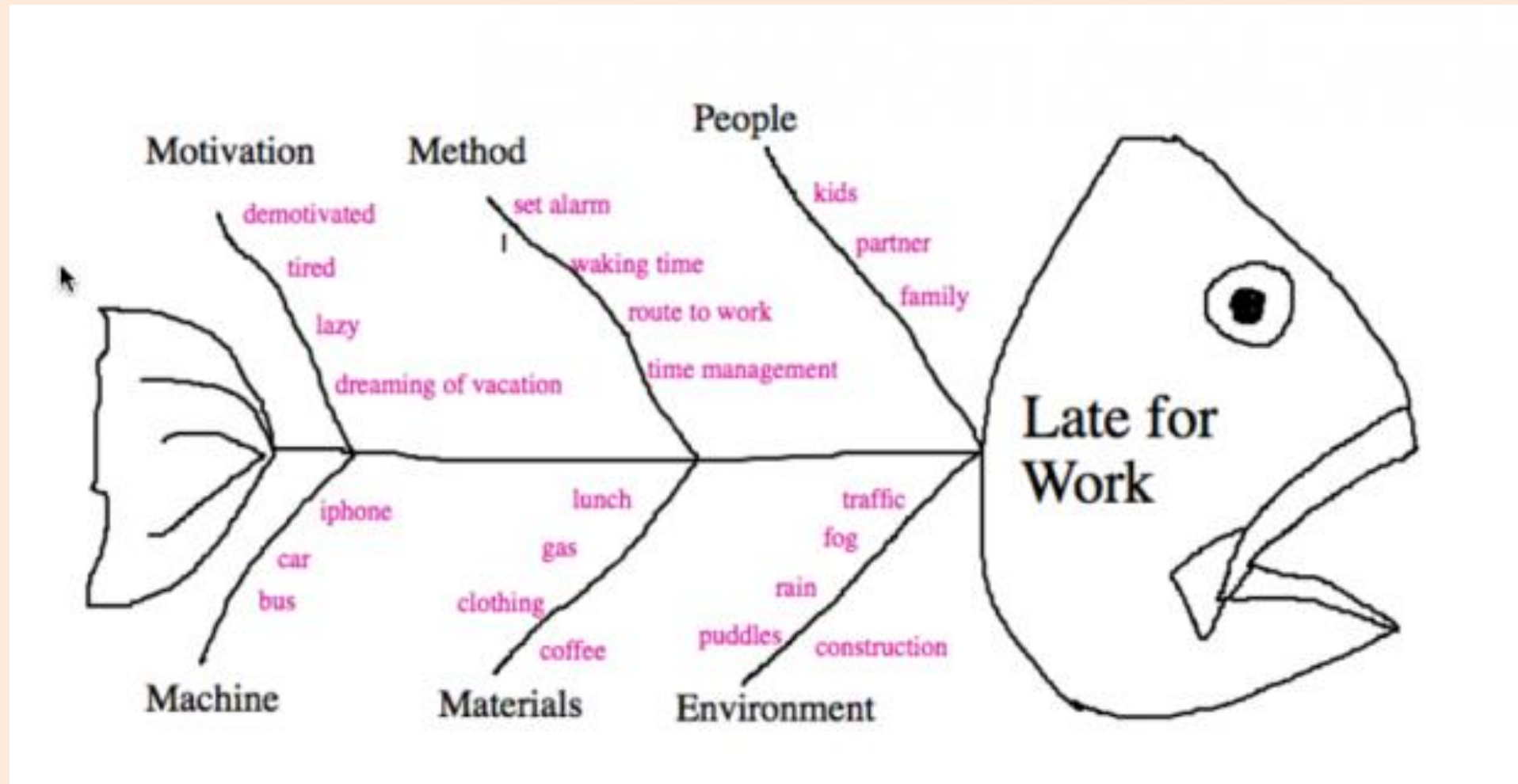
KAORU ISHIKAWA
1915 – 1989

Major Contributions

Japanese Organizational Theorist

- User Friendly Quality Control
- Fishbone Cause and Effect Diagram - Ishikawa diagram
- Implementation of Quality Circles
- Emphasized the Internal customer
- Shared Vision

Root Cause Possibilities





SHIGEO SHINGO
1909 - 1990

Major Contributions

Japanese Industrial Engineer

- Major contributor to the Toyota Production System now used by companies worldwide
- Also pioneered
 - “Just in time” production
 - “Lean” production



GENICHI TAGUCHI
1924 - 2012

Major Contributions

Japanese Engineer and Statistician

- Taguchi loss function, measures financial loss to society resulting from poor quality
- “*Off-line quality control*”- products and processes insensitive to parameters outside the design engineer's control
- Innovations in the statistical design of experiments,

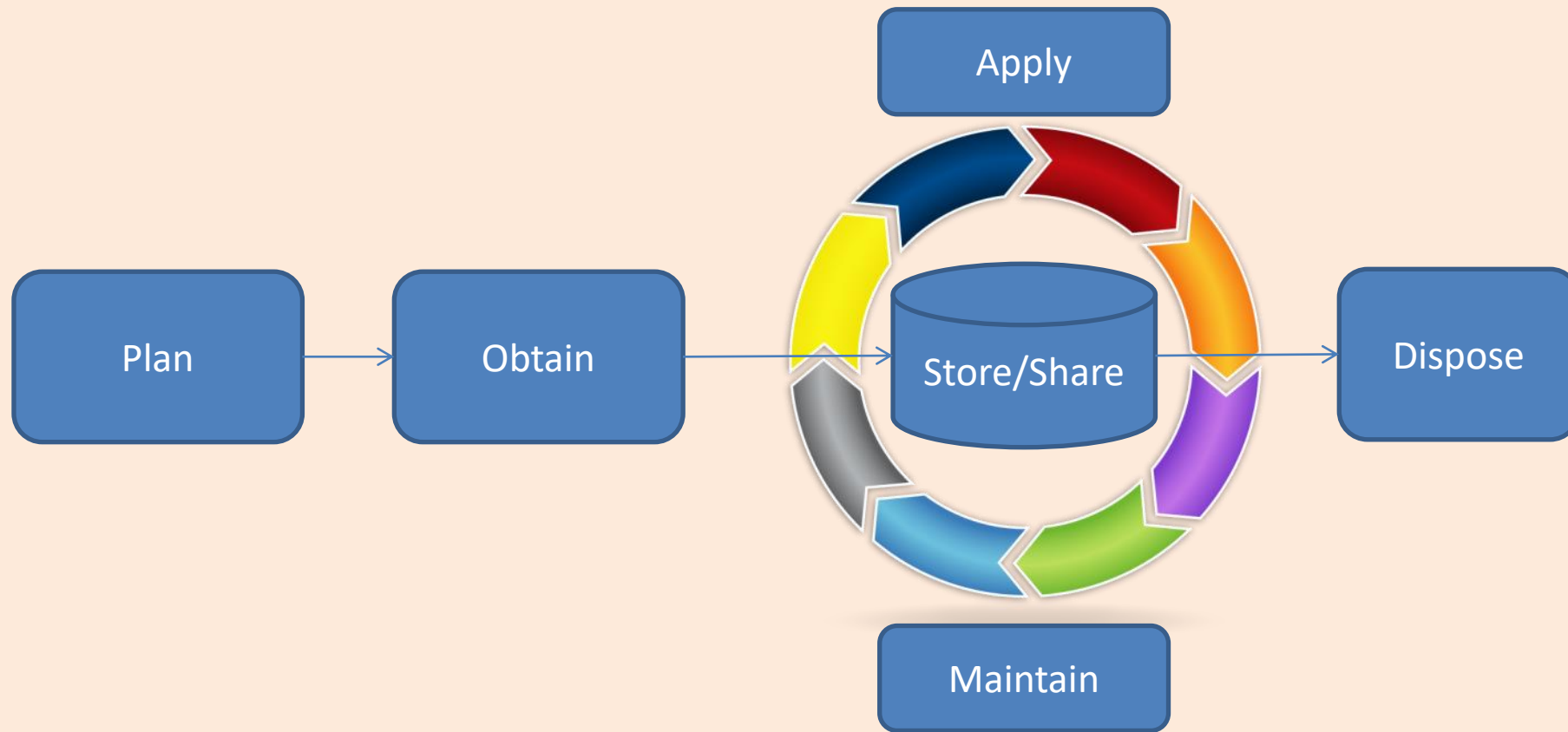
CONTEMPORARY ACADEMICIANS AND PRACTITIONERS



LARRY ENGLISH (1947-2020)

**FOUNDER OF INFORMATION IMPACT INTERNATIONAL
AND AUTHOR**

Non-Linear Life cycle



Books

- *Information Quality Applied: Best Practices for Improving Business Information, Processes, and Systems*
- *Improving Data Warehouse and Business Information Quality: Methods for Reducing Costs and Increasing Profits*



DR. RICHARD WANG

**DIRECTOR OF THE MIT CDOIQ PROGRAM &
UA LITTLE ROCK PROFESSOR OF INFORMATION SCIENCE**

Major Contributions

- Established the Total Data Quality Management (TDQM) Center at MIT
- Started the International Conference on Information Quality (ICIQ) in 1995
- Instrumental in establishing information quality as an academic discipline
- UA Little Rock Professor of Information Science and Executive Director of the UALR Institute for Chief Data Officers

Other Contributions

- Wang-Strong Framework of Data Quality Dimensions
- Developed the principle of “Managing Information as Product”
- Most cited author in information quality research publications
- Current research – Chief Data Officer

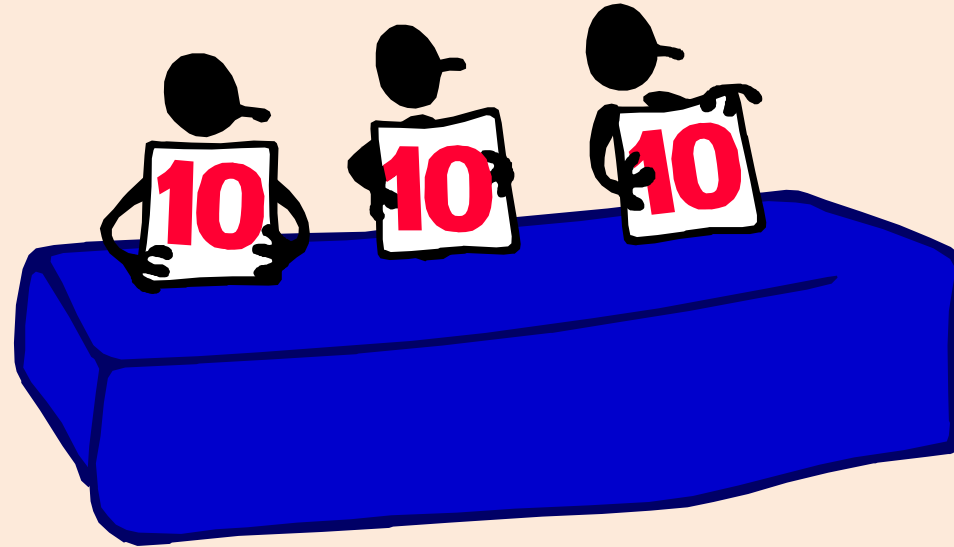
Books

- *Journey to Data Quality*, with Yang Lee, Leo Pipino, & James Funk
- *Quality Information and Knowledge*, with Kuan-Tsae Huang & Yang Lee
- *Introduction to Information Quality*, with Craig Fisher, Eitel Lauria, & Shobha Chengalur-Smith
- *Data Quality*, with Mostapha Ziad & Yang Lee



DR. THOMAS C REDMAN
AUTHOR AND CONSULTANT

Understanding Cost of Poor Data Quality



Redman's Rule of 10:

It costs 10x as much to complete a unit of work when input data are defective as it does when the input data are perfect

Information Chains



- An information chain is effectively a chain of processes through which information flows to achieve an objective in the organization.
- Only by understanding how information flows can you understand how the quality of the information
 - Affects the Organization
 - Is affected by the Organization

Books

- *Getting in Front on Data: Who Does What*
- *Data Driven: Profiting from Your Most Important Business Asset*
- *Data Quality for the Information Age*
- *Data Quality: The Field Guide*



DANETTE MCGILVRAY

AUTHOR AND CONSULTANT

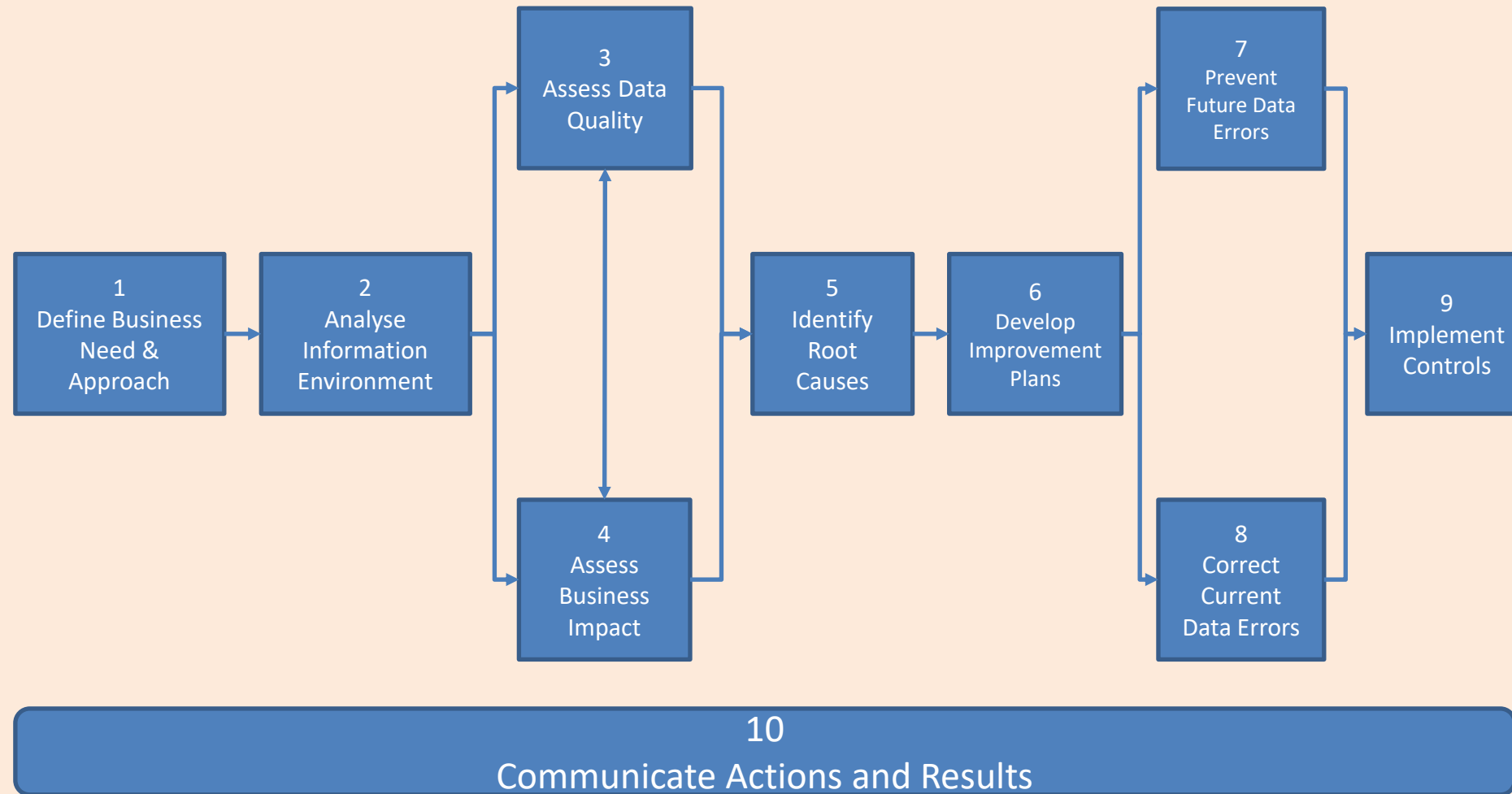
Books

- *Executing Data Quality Projects: Ten Steps to Quality Data and Trusted Information (1st Edition)*
 - Textbook for INFQ 7337
- *Executing Data Quality Projects: Ten Steps to Quality Data and Trusted Information (2nd Edition, New this year)*

IQ Culture and Governance Framework for Information Quality (FIQ)

Business Goals / Strategy / Issues / Opportunities (Why)						
	Plan	Obtain	Store and Share	Maintain	Apply	Dispose
Data (What)						
Processes (How)						
People & Orgs (Who)						
Technology (How)						
Location (Where) and Time (When and How Long)						
Requirements and Constraints:						
Responsibility:						
Improvement and Prevention:						
Structure and Meaning:						
Communication:						
Change:						
Culture & Environment						

The 10 Steps Method™





LAURA SEBASTIAN-COLEMAN

DATA QUALITY PRACTITIONER – AETNA, A CVS HEALTH CO.

Books

- *Measuring Data Quality for Ongoing Improvement: A Data Quality Assessment Framework (Morgan Kaufman)*
 - Textbook for INFQ 7303
- *Meeting the Challenges of Data Quality Management (in-process)*



DAVID LOSHIN
AUTHOR AND CONSULTANT

Books

- *Big Data Analytics (2013)*
- *Using Information to Develop a Culture of Customer Centricity (2013)*
- *The Practitioner's Guide to Data Quality Improvement (2010)*
- *Master Data Management (2008)*
- *Enterprise Knowledge Management (2001)*
- ...

Questions and Discussion